

1. Administrative & Study Identification

Full Study Title: A Phase I, Open-label, Multi-center Study to Evaluate the Safety, Tolerability, Dosimetry and Preliminary Activity of [177Lu]Lu-FF58 in Patients With Selected Advanced Solid Tumors **Short Title/Acronym:** A Phase I Study of [177Lu]Lu-FF58 in Patients With Advanced Solid Tumors **Protocol Number:** CAAA604A12101 **NCT Number:** NCT05977322 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [177Lu]Lu-FF58 (Radioligand Therapy) and [68Ga]Ga-FF58 (Radioligand Imaging Agent) **Phase of Development:** Phase I **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To test the safety, tolerability, and dosing of [177Lu]Lu-FF58 to identify the recommended dose and evaluate the occurrence of Dose Limiting Toxicities (DLTs). **Secondary Objectives:** To determine preliminary efficacy including response rate, Progression-Free Survival (PFS), Duration of Response (DOR), estimated effective dose by internal exposure evaluation, and further characterize the radioligand imaging agent [68Ga]Ga-FF58.

2.2 Background & Rationale To address the therapeutic need for targeted treatment in patients with advanced or metastatic tumors that express alpha-v beta-3 integrin ($\alpha_v\beta_3$) and alpha-v beta-5 integrin ($\alpha_v\beta_5$). The study aims to utilize a theranostic approach, using [68Ga]Ga-FF58 to screen and identify positive tumor lesions, followed by targeted radioligand therapy with [177Lu]Lu-FF58.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single Group Assignment (Single-arm) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 24 (Actual enrollment; the study was terminated early) **Number of Sites:** Multi-center (e.g., Israel, Netherlands, Spain, Switzerland) **Estimated Study Start:** 06-Oct-2023 (Actual) **Estimated Primary Completion:** 13-Dec-2024 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Patients with locally advanced unresectable or metastatic Pancreatic Ductal Adenocarcinoma (PDAC), locally advanced unresectable or metastatic Gastroesophageal Adenocarcinoma (GEA), or recurrent Glioblastoma Multiforme (GBM). 3. To be treated with [177Lu]Lu-FF58, patients must have at least one measurable lesion that shows [68Ga]Ga-FF58 uptake on PET/CT or PET/MRI.

4.2 Exclusion Criteria 1. Absolute neutrophil count (ANC) $< 1.5 \times 10^9/L$, hemoglobin < 10 g/dL, or platelet count $< 100 \times 10^9/L$. 2. Prior external beam radiation therapy (EBRT) to $> 25\%$ of the bone marrow, or creatinine clearance < 60 mL/min. 3. Non-GBM patients: Presence of symptomatic central nervous system (CNS) metastases, CNS metastases requiring local CNS-directed therapy (such as radiotherapy or surgery), or increasing doses of corticosteroids within 1 week before [177Lu]Lu-FF58 administration.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patients are screened with a [68Ga]Ga-FF58 PET/CT or PET/MRI scan. If tumor uptake is confirmed, varying doses of [177Lu]Lu-FF58 are administered (during the dose-escalation phase) every 3 or 6 weeks depending on the assigned schedule. **Route of Administration:** Intravenous infusion. **Duration of Treatment:** Treatment duration dictated by schedule, with a long-term follow-up period of up to 36 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care supportive therapies not expected to interfere with radioligand uptake. **Prohibited:** Increasing doses of corticosteroids within 1 week prior to [177Lu]Lu-FF58 administration (for non-GBM participants).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days prior to C1D1	Informed Consent, [68Ga]Ga-FF58 PET/CT or PET/MRI scan to confirm eligibility.
Baseline / Cycle 1	Cycle 1, Day 1	First Dose [177Lu]Lu-FF58, pre-infusion, end of infusion, and post-dose PK assessments.
Interim Assessments	C1 Days 2, 3, 4, 8	Radioligand therapy dosimetry and PK measurements at 24 hr, 48 hr, 72 hr, and 168 hr post-dose.
End of Study	Up to 36 Months	Disease progression monitoring, Long-term safety follow-up.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Occurrence of Dose Limiting Toxicities (DLTs) within the evaluation period (i.e., first 6 weeks starting from the first administration of [177Lu]Lu-FF58). **Measurement Method:** Any adverse event (AE) or abnormal laboratory value of CTCAE (v5.0) Grade 3 or higher that is not primarily related to disease progression.

7.2 Secondary & Exploratory Endpoints * Response Rate and Duration of Response (DOR). * Progression-Free Survival (PFS). * Estimated effective dose by internal exposure evaluation (Dosimetry parameters %IA/g).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Distribution of adverse events analyzed via frequencies for Treatment Emergent Adverse Events (TEAEs) and Serious Adverse Events (TESAEs). **Serious Adverse Events (SAEs):** Assessed continuously through monitoring of relevant clinical and laboratory safety parameters. **Data Monitoring Committee (DMC):** Internal and external safety review processes defined by Novartis Pharmaceuticals for Phase I dose-escalation trials.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety/Dose-Escalation Set (all patients who receive at least one dose of [177Lu]Lu-FF58). **Sample Size Justification:** Descriptive and exploratory sample size typical of a Phase I dose-finding study; target expanded to 24 enrolled participants before the study was ultimately terminated. **Handling of Missing Data:** Statistical analyses of the imaging properties and dosimetry are descriptive in nature, comprising summaries and graphical presentations. No formal hypothesis testing was performed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous site monitoring to capture all DLTs and safety metrics using CTCAE v5.0 parameters. **Audit Trail:** Standard robust electronic Case Report Form (eCRF) capture and source data verification managed by the sponsor.

11. Study Identifiers & Governance

Primary ID: CAAA604A12101 **Registry Number:** NCT05977322 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Phase I Study of [177Lu]Lu-FF58 in Patients With Advanced Solid Tumors **Official Regulatory Title:** A Phase I, Open-label, Multi-center Study to Evaluate the Safety, Tolerability, Dosimetry and Preliminary Activity of [177Lu]Lu-FF58 in Patients With Selected Advanced Solid Tumors

12. Clinical Framework (Oncology/Targeted Therapy Specific)

Primary Condition: Advanced Solid Tumors (PDAC, GEA, GBM) **Diagnosis Threshold:** Tumor lesions positive for [68Ga]Ga-FF58 uptake on PET/CT or PET/MRI **Medical Methodology:** Targeted Radioligand Therapy **Optimization Target:** alpha-v beta-3 integrin ($\alpha_v\beta_3$) and alpha-v beta-5 integrin ($\alpha_v\beta_5$)

13. Study Procedures & Methodology

Management Pathway: Diagnostic screening and radioligand dosimetry followed by therapeutic administration. **Theranostic Protocol (HCP):** Use of imaging agent [68Ga]Ga-FF58 to confirm therapeutic viability of [177Lu]Lu-FF58. **Patient Assessment:** Repeat dosimetry and pharmacokinetic measurements mapped against safety guidelines. **Quality Audit Mechanism:** Independent response evaluation utilizing RECIST 1.1 (non-GBM patients) or modified RANO (mRANO) criteria (GBM patients).

14. Observation & Follow-Up Schedule

Total Duration: Up to 36-month long-term follow-up period **Onsite Visit Cadence:** Intensive monitoring in Cycle 1 (Days 1, 2, 3, 4, 8) with 3-week or 6-week cycles thereafter **Safety Monitoring Frequency:** Continuous throughout treatment and long-term follow-up **Efficacy Review Interval:** Per RECIST/mRANO scheduled assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				